A tool's registers: the daily record and the outcomes

Above: TOOL.md §3.4 (record) and §3.5 (measure).

Talks to:

- the card, whose cards the photo freezes;
- the screeners, for the capture minute, the entry minutes and the screener versions;
- the market reader, whose history R2 is a view of;
- the data sources, for the outcome bars;
- the scorecard, which trains and judges on R4;
- qp, which backtests over R1 and reads R3, and the shared record, which holds a copy.

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1. What it is

How a day of screening becomes data that can be judged, learned from and backtested, without look-ahead. Each tool keeps its own registers — tools are separate experiments — and every value in them is exactly what was known at its minute.

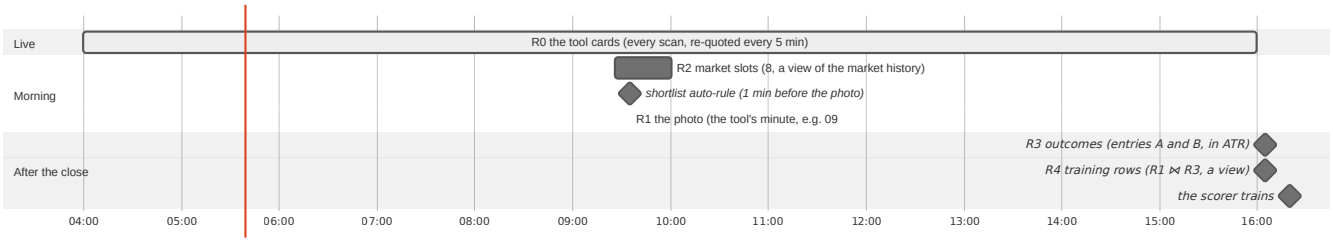
Register	The question it answers
R0	what is on screen right now?
R1	what did each card look like at the tool's photo minute?
R2	what was the market doing around the photo?
R3 (A / B)	from entry A / B, how far did each photographed card run, each way, in ATR?
R4 (A / B)	the photo and its outcome, side by side: one example per card per day
Shortlist	what did the trader, or the auto-rule, pick?

What it owns: the photos (R1), the outcomes (R3) and the shortlist, in the tool's own file, with a copy in the shared record.

What it never does: take a photo late and call it on time, guess a missing bar, or carry a value that was not known at the minute it records.

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2. The diagram



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3. The registers, in full

3.1 R0 — the live list, as rows

R0 is the day's **tool cards**: live (matched this scan) and stale (matched earlier, still re-quoted every 5 minutes). In the design they are **rows saved to the tool's file after every change**, not a memory list with a checkpoint, so a restart loses nothing and the day's list is kept after the close. Cleared at midnight.

3.2 R1 — the photo

At the tool's capture minute, the register job writes, for **every tool card of the day** (live or stale):

Column	Contents
`tool` , `day` , `ticker`	the key
`captured_at`	the instant
`card`	the whole card as JSON: every fact family with its value, status, source and `known_at`
`tool_card`	the whole tool card: screener keys **with their versions and first-match times**, first seen, live, star, manual bias, score (with model version, sample, confidence)
`market_state_id`	the market state the card carried
`capture_version`	the version of the photo's definition

Nothing is left out, so any column can be added to an export later without losing a past day. The exported column set carries forward the legacy photo — identity, price, moving averages, volatility and volume, the four range windows, size, short interest, pre-market, classification, market context, the 27 relations, the 31 CANSLIM columns, the catalyst label, the cross-tool flag, and `foundMinsFromOpen` — the exact list in legacy/screeners/REGISTERS.md.

One photo, at its minute. Only the scheduler takes a photo, at the capture minute. A photo that could not be taken then (the desk was down) is recorded as **missed**, with the reason. A photo taken later by hand keeps its real instant and is marked **late**; late photos stay out of training and verdicts unless included on purpose, because they hold what was known after the method's minute. **A photo is never replaced** — the old desk's `POST /api/warehouse/collect` re-ran the capture at any time and silently overwrote the day (D-D8).

3.3 R2 — a view of the market history

R2 is the market around the photo, at eight fixed slots (09:26, 09:31 … 10:01). In the design it is **generated from the market states** the market reader stored (MARKET.md), so it carries the O'Neil status **as it was** in that state — not re-read at export time, where an old day used to show today's status (D-D3). Its columns (SPY/QQQ/IWM/DIA/VIX, the desk read, the O'Neil status, and the 15 sectors) are those in the legacy register.

3.4 R3 — outcomes, on the trading calendar

At 16:05 on each trading day (half days: 15 minutes after their close), for every R1 row:

Step	Rule
bars	1-minute and daily bars through the bar adapters (Yahoo, then Alpaca), stored **once per ticker per day** and shared by every tool
ATR14	from the 14 sessions **before** the day, on the trading calendar — no look-ahead, and no weekday arithmetic that treats a weekend as missing data
entries	the **open** of the bar at each of the tool's entry minutes (A and B)
hh, ll	from the entry bar to the close: highest high, lowest low
upR, downR	`upR = (hh − entry) / ATR14`, `downR = (entry − ll) / ATR14`, both positive
opening range	09:30–09:35 high and low
result	`ok`, or a **reason row** (`no bar at 09:37`, `no daily history`, `halted`) — every ticker not measured says why, rather than being dropped with only a log line

A **good move is `upR ≥ 1.3`** — the same line everywhere. There is no stop, target or exit: this measures what a tool finds, not a trade. R3 is **one table with an `entry` column** (`A`, `B`), so a third entry can be added without a new table. Worked example: entry $10, high $12.50, low $9.40, ATR $1 → upR 2.5, downR 0.6.

3.5 R4 — a view, not a copy

`R4 = R1 ⋈ R3` on tool, day, ticker and entry. **Every R1 column is available** — CANSLIM, the wider ranges, all of it — because R4 is a view, not a narrower copied table that drifts from R1 (D-D2). A model or the scorecard selects the columns it uses. **Uploaded rows** (examples from outside the tool) live in their own table, labelled by source, and are excluded from verdicts unless explicitly included, so a verdict is never diluted by outside data.

3.6 Shortlist

One row per tool, day and ticker: `added_at`, `method` (manual, auto), and the card's price, change, sector and score at that moment, with `exported_at` when it was copied to TradingView. The **auto-rule** runs one minute before each tool's photo, adding the top N cards above a minimum score — which now have a score, where with the scorer off the old rule added nothing (D-D1). The **unified shortlist** is a query over every tool's table (no shared file rewritten by many processes), and the card's ☆/★ is read from here, so a pick is recorded once.

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4. Contracts

It owns (the tool's file, with a copy in the shared record)	Key
`r1_photos`	tool, day, ticker
`r2_slots` (a view over the market history)	tool, day, slot
`r3_outcomes`	tool, day, ticker, entry
`r4` (a view, R1 ⋈ R3)	tool, day, ticker, entry
`r4_uploads`	tool, day, ticker, entry, source
`shortlist`	tool, day, ticker
`bars_day` (shared outcome bars)	ticker, day

It reads	From
the capture minute, the entries, the screener versions	the tool and its screeners (SCREENERS.md)
the cards and tool cards	the card pipeline (CARD.md)
the market states	the market reader (MARKET.md)
the outcome bars	the bar sources (DATA-SOURCES.md)

Read by	What
the scorecard	R4 per tool (LEARNING.md)
qp	R1 per tool and day (backtest universes) and R3 (outcomes), as of a day (BACKTEST.md)
the journal	the R1 row known at a trade's entry (REVIEW.md)

Exports are generated from the schema, never a hand-kept column list, so a stored column is never missing from an export (D-D4): CSV or JSON per register and day on the Warehouse tab and the API; the nightly analysis set (Parquet, and CSV for old notebooks); and the legacy CSV layout, with `price936`, on request — the photo's price is `price` now, and the historical name lives only in that legacy export (P-1).

5. When it fails

Event	What happens
the desk was down at the capture minute	the photo is recorded as **missed**, with the reason; never taken late as if on time
a photo is taken by hand later	it keeps its real instant and is marked **late**; out of training and verdicts unless included on purpose
a ticker has no bar at an entry minute	a **reason row** says so (`no bar at 09:37`); it is not silently dropped
a daily history is too short for ATR14	the outcome row's result names it; no guessed ATR
the shared record is down	the register is written to the tool's file; the copy catches up later

6. Built on

The old desk: every register and column in legacy/screeners/REGISTERS.md, the photo and the outcome measure in legacy/screeners/METHODOLOGY.md.

What failed there:

ID	Fault	Prevented by
D-D8	a manual route re-ran the photo at any time and silently replaced the day	only the scheduler takes a photo; a late one is marked late, never replaced
D-D3	an old R2 day showed today's O'Neil status	R2 is a view of the market history, which holds M as it was
D-D2	R4 was a copied table narrower than R1	R4 is a view over R1 and R3
D-D4	the opening range was stored but not exported	exports generated from the schema
P-1	`price936` whatever the capture time	the price is `price`; the legacy name only in the legacy export
D-D1	the auto-rule added nothing (no scores)	scores computed in process (LEARNING.md)
—	outcomes on weekday arithmetic; a missing bar dropped a ticker with only a log line	the trading calendar; a reason row for every ticker not measured
—	registers in one SQLite file per tool, read by qp over HTTP from each port	a copy in the shared record; qp reads it as of a day

7. Status

Part	Status
R1 photos (stamped with their screener versions), R3 outcomes and the shortlist dual-written to the shared record; all eleven tools' history imported with 0 differences	**built, in shadow** (the tool's file stays the record until the switch)
R0 as rows, R4 as a view, R2 as a view of the market history, reason rows for every ticker not measured	**to build**
the switch to read registers from the shared record	**to build** (step D; BACKTEST.md)